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Enhanced MobileNet for skin cancer image classification with fused spatial channel attention mechanism

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Skin Cancer, which leads to a large number of deaths annually, has been extensively considered as the most lethal tumor around the world. Accurate detection of skin cancer in its early stage can significantly raise the survival rate of patients and reduce the burden on public health. Currently, the diagnosis of skin cancer relies heavily on human visual system for screening and dermoscopy. However, manual inspection is laborious, time-consuming, and error-prone. In consequence, the development of an automatic machine vision algorithm for skin cancer classification turns into imperative. Various machine learning techniques have been presented for the last few years. Although these methods have yielded promising outcome in skin cancer detection and recognition, there is still a certain gap in machine learning algorithms and clinical applications. To enhance the performance of classification, this study proposes a novel deep learning model for discriminating clinical skin cancer images. The proposed model incorporates a convolutional neural network for extracting local receptive field information and a novel attention mechanism for revealing the global associations within an image. Experimental results of the proposed approach demonstrate its superiority over the state-of-the-art algorithms on the publicly available dataset International Skin Imaging Collaboration 2019 (ISIC-2019) in terms of Precision, Recall, F1-score. From the experimental results, it can be observed that the proposed approach is a potentially valuable instrument for skin cancer classification.

Keywords Skin cancer, Classification, Deep learning, Machine vision, Medical image analysis

Skin cancer is one of the most lethal cancers globally[1]. Among different types of skin cancers, Melanoma is considered the most brutal type of skin lesion with the highest death rate annually[2]. Invasive melanoma accounts for only 1% of cases of skin cancer, but a vast majority of skin cancer deaths. In 2023, an estimated 97,610 new cases of invasive and 89,070 cases of in situ melanoma will be diagnosed in the US, while 7,990 people will die from this disease. Early detection of skin cancer like melanoma is vital to clinical diagnosis and treatment. The common way to implement skin cancer detection early is to periodic exterminate the changing or growths of skin lesions, particularly the ones that render unusual[3]. New lesions or the progressive change in the lesion's size, shape, and color could be identified by the clinicians promptly[4,5]. However, manually visual inspection is time-consuming, laborious, and error-prone[6]. The accuracy of manually visual diagnosis by experienced dermatologists is no more than 60%[7]. In addition, there is a shortage of radiologists and related experts that are aware of skin cancer screening in underdeveloped and developing countries[8]. To summarize, it is crucial to implement an automatic machine vision-aided system to facilitate the dermatologists to make better decision in clinical practice.

On one hand, the machine learning-based techniques can realize automated screening and diagnosis of skin cancer. For instance, the work of Ref.[9] presented a software platform for automatic detection of melanoma in dermoscopic images. To achieve this purpose, the proposed system employed a group of image processing techniques and three supervised-learning applications. In 2020, Fei et al.[10] presented a classification framework using an algorithm with support vector machine (SVM) machine learning and the particle swarm optimization (PSO) principle for improving the performance of skin lesion in clinical diagnosis. The Hospital Pedro Hispano dataset containing 200 images was leveraged in this study. In total, 46 texture features were incorporated to complete the skin cancer image analysis and classification. To classify the melanoma skin cancer using the

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static features, including geometry, color, and texture in the images, Hurtado and Reales[11] proposed a skin cancer classification approach by using a digital camera and data augmentation technique using the smoothed bootstrapping algorithm. In general, eight different classifiers with various topologies, such as k-nearest neighbor (KNN), artificial neural network (ANN), and SVM, were included in the comparison experiments. And the ANN classifier achieved an area under curve (AUC) of 87.1%. To avoid the expensive diagnosis and assist the dermatologists, Javaid et al.[12] presented a skin lesion classification method (benign and malignant) using both image processing and machine learning. First of all, the OTSU thresholding method was exploited. Then, the gray level co-occurrence matrix (GLCM), histogram of oriented gradients (HOG), and color identification features were extracted from the pre-processed images. In addition, the principal component analysis (PCA) was used to reduce the dimensionality of the HOG features. Finally, the quadratic discriminant, SVM, and random forest (RF) were taken as classifiers. In 2022, in the work of Ref.[13], Karitha, Jayalakshmi, and Kamalakkannan presented a detection method for malignant and non-cancerous cells with image processing methods. A group of machine learning classifiers, including Naive Bayes (NB), RF, KNN, and SVM were exploited. With the SVM classifier, it achieved a classification accuracy of 97.7%.

On the other hand, the deep learning models, which rely heavily on large number of training samples, have been exploited in a variety of applications[14–17] and skin cancer classification. To note that the convolutional operators are commonly exploited for extracting the local features from the skin lesion images. Due to the similar appearance between malignant and benign skin lesions, Ref.[18] proposed a skin cancer classification method using the convolutional neural network (CNN). In this CNN architecture, there are three hidden layers with the output channels of 16,32, and 64, respectively. Meanwhile, this model employs a set of optimizers, including SGD, RMSprop, Adam, and Nadam with the learning rate as 0.001. At last, the Adam optimizer generated the optimal accuracy of 99% for classifying the skin lesions of four classes, which consists of dermatofibroma, nevus pigmentosus, squamous cell carcinoma, and melanoma. Sharma et al.[19] proposed a cascaded ensembled network based on a CNN and a multi-layer perceptron model. Generally, the accuracy of ensembled deep learning model was enhanced from 85.3 to 98.3%. To be specific, the presented model used the CNN to reveal the non-handcrafted features rather than the handcrafted ones such as color and texture. Recently, Atta et al.[20] presented a CNN model for addressing the skin cancer classifications using 3,600 images with the resolution of 224×224 . Furthermore, Endah[21] also proposed a CNN-based approach for classifying the benign and malignant skin cancer images. In 2023, Mukadam and Patil[22] realized a custom CNN model on the publicly available dataset HAM10000. It achieved the accuracy of 98.77%, 98.36%, and 98.89%, for protocol-I, II, and III, respectively.

Receptive field is used to represent the receptive range of neurons at different positions in the images[23]. The reason why neurons are unable to perceive the global information in an image is that the convolutional[24] and pooling[25] units are commonly used in the networks. Accordingly, the self-attention mechanism has emerged from the domain of deep learning. Following the early work of attention mechanism[26], recent works in the field of computer vision have began to leverage transformers, e.g. the vision transformer (ViT)[27]. Accordingly, ViT has been applied in skin cancer classification due to its characteristics. For instance, a two-stage pipeline for the categorization of skin cancer was proposed by Aladhadh et al. in their article[28]. To boost the data samples in the first step, a collection of data augmentation operations was utilized. And by separating the input images into consecutive patches, the ViT model designed for medical images was utilized. Finally, the classification result was generated using the multi-layer perceptron (MLP) module. The application of ViT for skin cancer categorization was then assessed by Nikitin and Shapoval in the study[29]. Arshed et al.[30] suggested a method for categorizing various forms of skin cancers based on various attention mechanisms.

This study proposes a novel deep learning model to implement skin cancer classification, inspired by the MobileNet model[31] and named after MobileNet-MFS, in which MFS denotes **M**ultiple **F**used **S**patial-channel attention. In the proposed model, the self-attention mechanism is incorporated with the convolutional modules to manipulate both the local receptive field and the global associations between long-range pixels. In the following tests, a public dataset[32–34] with eight classes of skin lesions is used to evaluate the performance of the proposed approach. Finally, comparative studies between the proposed strategy and cutting-edge techniques were carried out. Experimental findings show that the suggested technique outperforms the state-of-the-art utilizing a variety of assessment metrics.

In general, the contributions of this work can be summarized as:

- This study proposes a novel deep learning model based on MobileNet for skin cancer image classification.
- A novel attention mechanism is proposed, namely the fused spatial channel attention, which is used to take the place of Squeeze-and-Excitation (SE) module in the original MobileNet v3 model.
- Results from experiments show that the presented technique performs better than the state-of-the-arts.

Results

Implementation details and evaluation metrics

The experiments detailed below were conducted utilizing four NVIDIA RTX 3090 graphical processing units (GPUs) and the Pytorch platform[35]. The skin cancer images are then resized to a consistent resolution of 224 pixels on both the width and height dimensions. In addition, a batch size of 16 was employed, the Adam Optimizer was utilized, the initial learning rate was set to 1e−4, and the number of epochs was established as 100.

The experiments contained many assessment criteria, namely accuracy, precision, recall, F1-score, and AUC, to evaluate the performance of the proposed model and the comparison methodologies.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

where TP, TN, FP, and FN represent the counts of true positives, true negatives, false positives, and false negatives, respectively.

Experimental results of the proposed approach

The accuracy curves of the suggested technique are presented in Fig. 1. It is evident that, following 20 epochs, the accuracy of the proposed model has surpassed 90% on the training set and beyond 80% on the testing set.

Furthermore, the loss curves are depicted in Fig. 2. Based on the results obtained, it can be inferred that the suggested approach exhibits a loss of around 0.6 on the training set after 100 epochs, and a loss of approximately 0.5 on the testing set.

The confusion matrix depicting the performance of the proposed technique on the eight categories of skin cancer photos in the ISIC-2019 dataset is illustrated in Fig. 3. It is important to acknowledge that the values within the confusion matrix have been appropriately normalised. It is important to acknowledge that the numerical values adjacent to the left column and bottom rows of the confusion matrix represent the many categories of skin cancer photos. In particular, the symbols 0, 1, 2, 3, 4, 5, 6, 7 represent the following entities: AK, BCC, BKL, DF, MEL, NV, SCC, and VASC, respectively. The results indicate that the BCC and NV pictures have demonstrated the highest level of predictive accuracy when employing the proposed methodology.

The proposed MobileNet MFS achieved over 80% accuracy for BCC and NV classifications. However, the performance for MEL was less robust. This can be attributed to the following reasons.

- Class imbalance: It is a common issue in medical image classification datasets. The uneven distribution of samples across different classes can lead to models performing well on the majority classes while neglecting the minority classes.
- Intra-class and inter-class similarity: The difficulty of classification is directly related to the similarity of instances between classes. Classes with high intra-class variability and low inter-class separability, such as MEL, are inherently more challenging to classify.
- Data complexity: The complexity of the data, including the scale and resolution range, can affect model accuracy. MEL, being one of the most aggressive forms of skin cancer, often presents with complex and varied patterns that are challenging for the model to learn from.

The receiver operating characteristic (ROC) curves, which depict each type of skin cancer image, are presented in Fig. 4. The suggested technique demonstrates a notable achievement in striking a potential equilibrium between the true positive rate and the false positive rate across the majority of skin cancer images, with the exception of the MEL category.

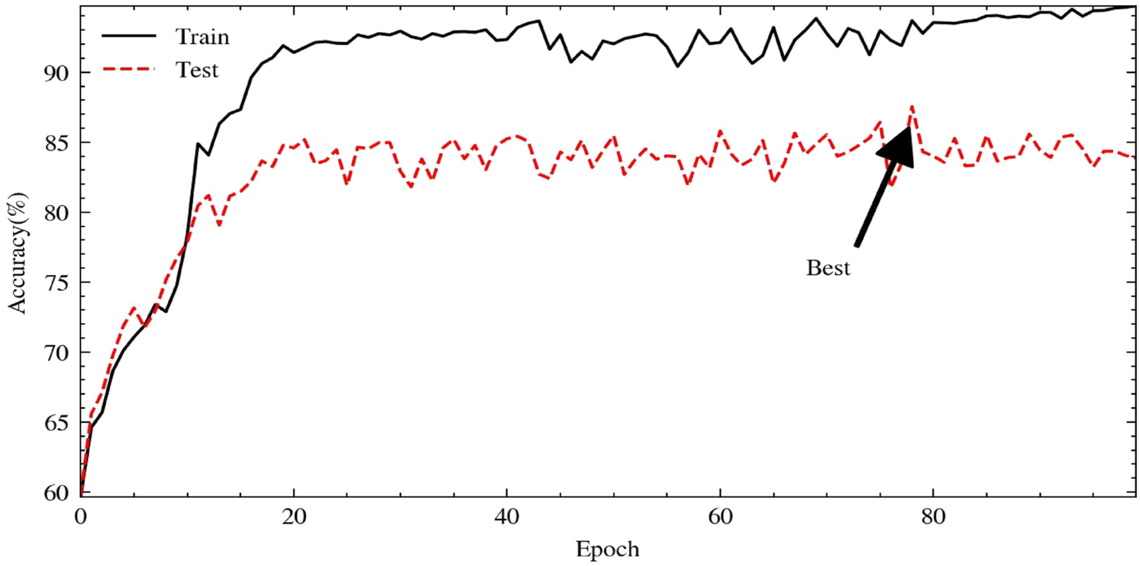

Fig. 1. The accuracy curves of the proposed approach on the ISIC-2019 dataset.

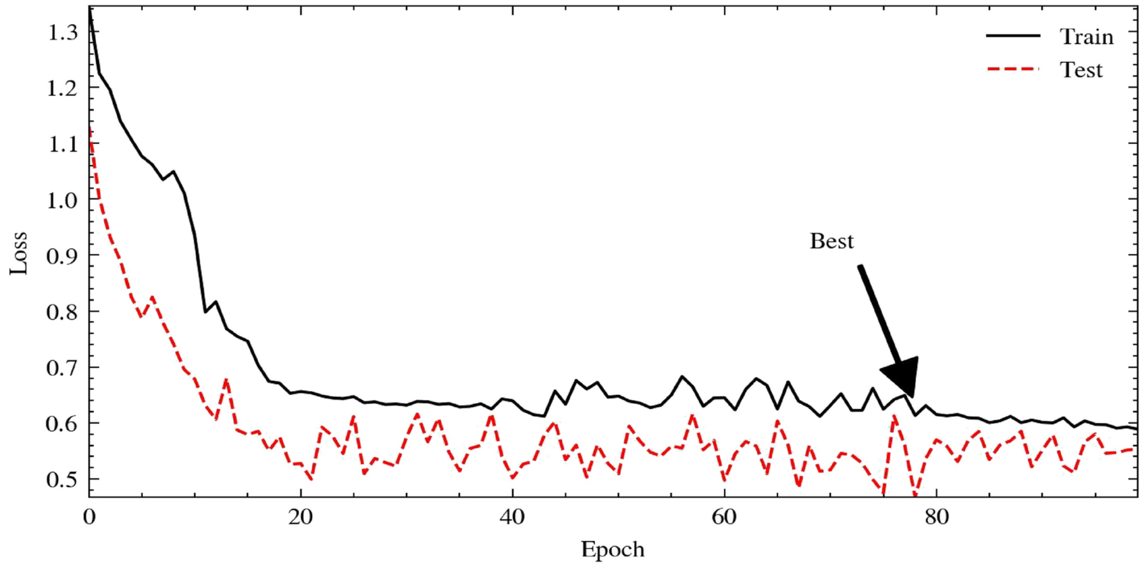

Fig. 2. The loss curves of the proposed approach on the ISIC-2019 dataset.

Comparison experiments

Comparison between various attention mechanisms on the proposed deep learning model

To visually represent the effects of various attention mechanisms, we conducted a comparative analysis of the accuracy achieved by several attention mechanisms (SE[36], ECA[37], CBAM[38], CA[39], FSCA[40], MFS (The proposed attention mechanism)) within the presented MobileNet v3 framework. As depicted in Fig. 5, the FSCA attention mechanism and the combined multi-scale MFS attention mechanism exhibit notable increases in growth over epochs. In addition, when the epoch is increased to 20, the maximum value of the system exhibit optimal performance. In contrast, the amplitude of variation in other attention mechanisms is comparatively considerable, whereas the MFS and FSCA processes exhibit the highest level of precision.

Comparison between the state-of-the-arts and the proposed approach

The comparative analysis also includes an evaluation of the accuracy of various deep learning models and the proposed MobileNet-MFS model. The accuracy curve of MobileNet-MFS is depicted by the light blue curve in Fig. 6. In comparison to other models, it exhibits a rapid ascent and progressively attains its high-level platform during a span of 20 epochs. In conclusion, upon reaching epoch 78, the proposed MobileNet-MFS model achieves a peak accuracy above **87.3%**, hence outperforming the competing models.

To conduct a complete comparison between our model and other established models, we also performed calculations on many metrics including precision, recall, F1-score, and AUC. These indicators have the ability to assess the capabilities of the competing methods from many perspectives. Based on the data shown in Table 1, it is evident that the MobileNet-MFS model exhibits superior performance in terms of precision, recall, and F1-score. Nevertheless, the MobileViT model exhibits a superior AUC value compared to the proposed technique.

When evaluating the performance of models, it is imperative to take into account the computing resources utilized by the models, in addition to the aforementioned indicators. MobileNet-MFS is derived from MobileNet v3[31] and is classified as a lightweight CNN. The model's low weight facilitates its use across a broader spectrum of scenarios. Furthermore, the computational complexity of the model is a crucial determinant, and the Floating Point Operations (FLOPs) serve as a reliable metric for quantifying the computational complexity of the model. The indicators presented in Table 2 provide a comprehensive assessment of multiple dimensions of the model. When considering the number of parameters, memory size, and FLOPs counts, it can be observed that the MobileNet-MFS offers advantages compared to ResNet34[42] and DenseNet-121[24] in terms of the number of weighting parameters. Although it consumes slightly more processing resources than MobileNet v3, it is not as streamlined as ShuffleNet v2[41] and ResNet34.

Moreover, to illustrate the performance of the proposed approach on larger scale image dataset, we further conducted comparison experiments between the state-of-the-art attention mechanism and deep learning methods and ours on the mini-ImageNet dataset[43], which contains 100 categories of 60,000 color images with 600 images per category. The experimental results are provided in Tables 3 and 4.

It can be observed that the proposed approach has achieved superior performance over the state-of-the-art methods on mini-ImageNet in terms of Precision, Recall, and F1-score. To note that the performance of SE mechanism in Table 3 and the peroformance of MobileNet v3 in Table 4 are exactly the same because the original MobileNet v3 model leverages the SE mechanism.

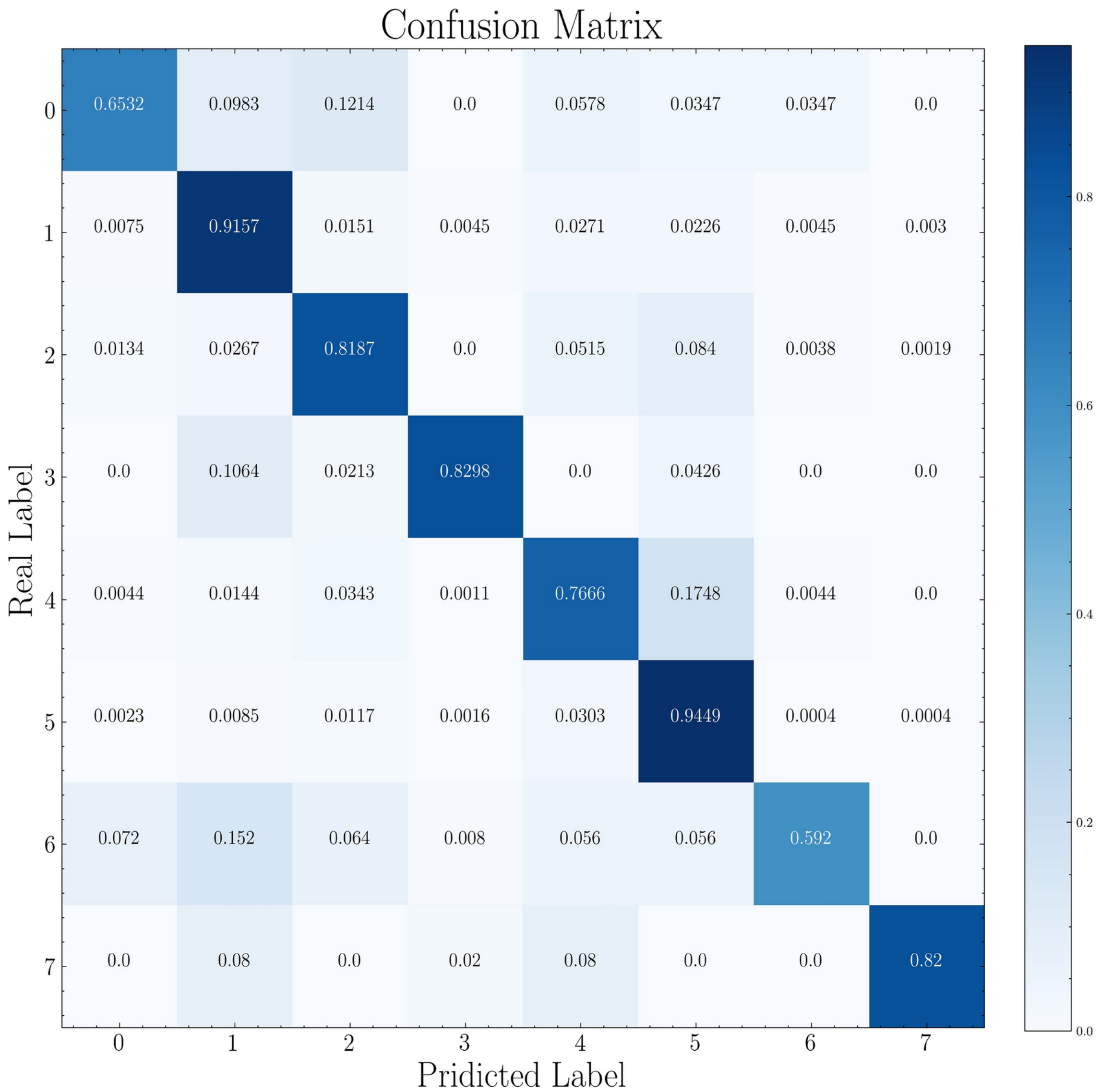

Fig. 3. The confusion matrix of the proposed approach on the ISIC-2019 dataset. (Eight different types of skin cancer images).

Discussion

The majority of CNN-based deep learning models are able to extract feature maps from images; if these models adopted a deeper network structure, they would be much better at extracting features from images. However, because the convolutional module concentrates on the local receptive fields of images, the performance of CNNs might be constrained. This results in the ignoring of the dependencies that exist between the pixels in an image that are separated by considerable distances. Meanwhile, improving deeper CNN models implies an increase in the amount of processing resources available.

In the images of skin cancer, the regions that contain the lesions are often dispersed over the whole image rather than being confined to one particular area. Meanwhile, the addition of extra layers to the CNN models would not always ensure improved image classification performance due to the fact that the mechanism of the local receptive field is unable to deal with this type of images. As a consequence, a self-attention mechanism-based skin cancer image classification model is presented to make advantage of the relationships between the long-range pixels observed throughout the images. In particular, the suggested approach constantly extracts the correlation between image patches by utilizing the mechanism of MSA. Consequently, the proposed method is able to preserve the information that is helpful for classification. The provided approach, is able to extract valuable information from images while simultaneously lowering the amount of computer resources it requires. Nevertheless, there are several drawbacks to this study: the inconsistency of the picture samples in the dataset that was utilized in the trials was a limitation that placed a damper on the performance of the suggested method; the attention mechanism used in the proposed model still needs large amount of computing resources to function properly.

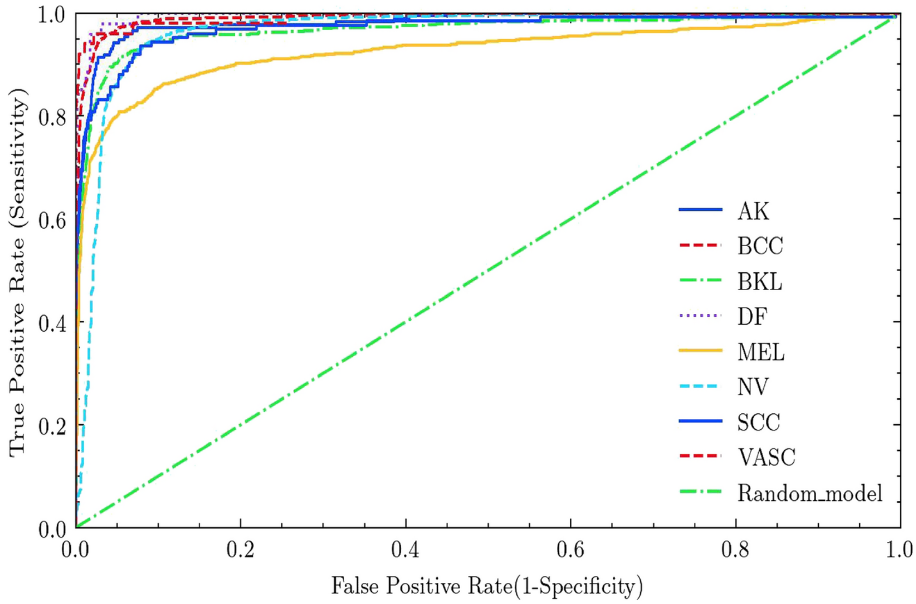

Fig. 4. The ROC curve of the proposed approach on the ISIC-2019 dataset (Eight different types of skin cancer images).

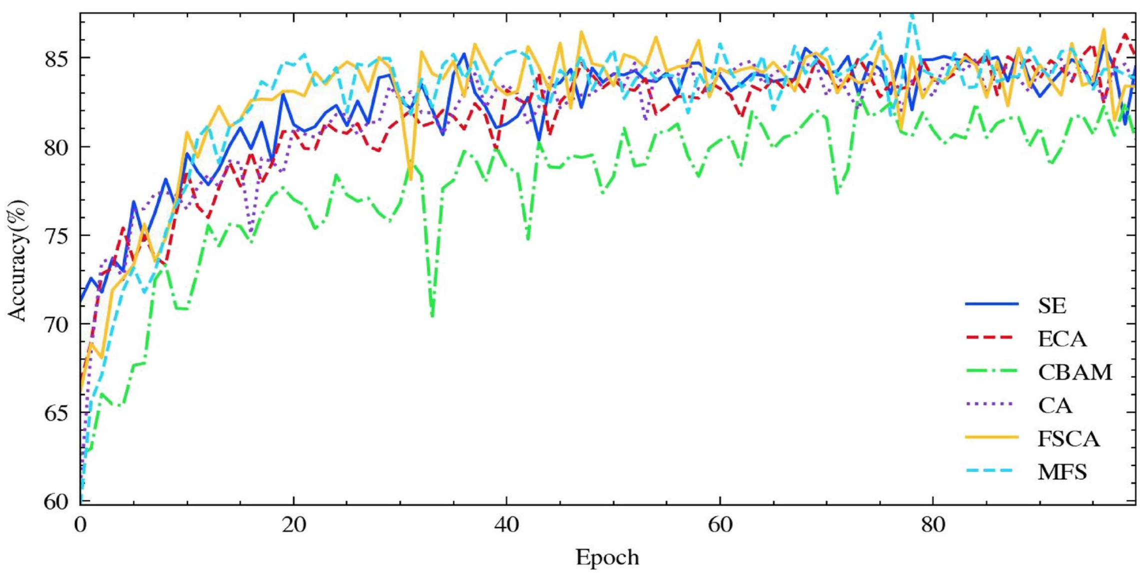

Fig. 5. The influence of various attention mechanisms on the proposed approach for classifying the images in ISIC-2019 dataset.

In addition, as demonstrated in Fig. 7, the proposed approach has also generated misclassified outcomes. As shown in Fig. 7, the image of BCC category is wrongly classified into the class of NV. It can be observed that the misclassified case could be attributed to the close similarity between the ground truth and the predicted outcome. We have used class activation mapping techniques to visualize the regions of interest that the model focuses on during prediction. This has helped to identify that certain misclassifications, such as BCC being misclassified as NV, may be due to similar textures or color patterns in the images of these classes. For instance, the model might be focusing on the pigmented areas in BCC lesions that resemble the seborrheic keratosis patterns in NV cases.

The purpose of this work is to categorize images of skin cancer using a network architecture that is based on both CNN and self-attention mechanism. When compared to the current state-of-the-art techniques, the performance of the proposed model was found to be superior. The efficiency of the suggested model is demonstrated by experimental findings based on the dataset.

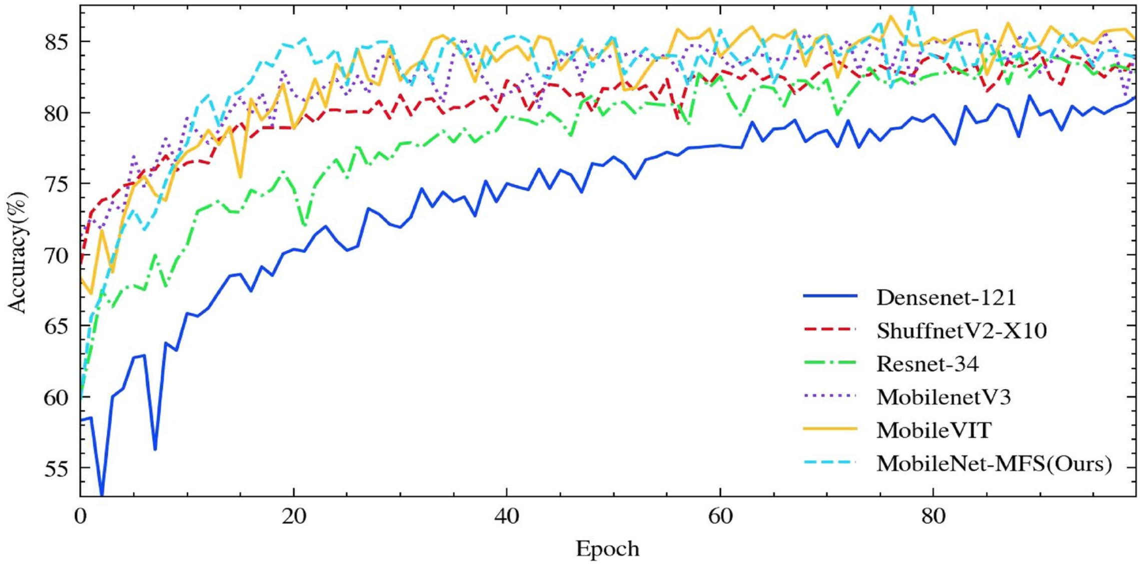

Fig. 6. Performance comparison between the state-of-the-art attention mechanisms and the proposed approach on the ISIC-2019 dataset.

Method	Precision	Recall	F1-score	AUC
MobileNet v3	0.8601	0.8574	0.8568	0.9642
Desnenet 121	0.8053	0.8107	0.8051	0.9501
ShufflenetV2_X10[41]	0.8407	0.8429	0.8389	0.9595
Resnet34	0.8412	0.8447	0.8403	0.9615
MobileViT	0.8669	0.8660	0.8662	**0.9697**
Our work	**0.8732**	**0.8751**	**0.8730**	0.9631

Table 1. Comparative results of the current state-of-the-arts and the proposed method on the ISIC-2019 dataset. Significant values are in bold.

Model	Top-1 Acc. (%)	Parameters (M)	Memory (MB)	FLOPs
MobileNet V3	85.70	4.21	50.39	226.44
DenseNet 121	81.17	6.96	147.10	2881.60
ShuffleNet V2_X10	84.26	1.26	20.84	149.58
ResNet34	84.47	21.29	37.61	3673.72
MobileNet-MFS	87.53	4.96	51.30	254.94

Table 2. Further comparison between the competing methods. Acc. denotes Accuracy and M represents millions.

In recent times, various deep learning models based on self-attention mechanism have demonstrated promising performance in machine vision tasks. As a result, in the not-too-distant future, we will conduct more research into the multi-model and multi-label deep learning models for the classification and prediction of skin cancer. Moreover, since the AUC performance of MobileViT model has been proven more promising than the proposed approach on the same dataset, we will also delve into the applications of vision transformer models in medical image analysis.

Methods

Due to the absence of biological pattern information, many classification techniques mainly rely on manual feature sets, which have a constrained capacity to generalize to dermoscopic skin images. Various skin lesions are closely matched in terms of size, shape, and color, making them heavily connected and giving poor feature information. Therefore, it is futile to classify skin using manual feature-based approaches. On the other hand, the deep learning-based algorithms aim at automatically extracting the most optimal features. In comparison to shallow networks, deep networks especially the CNN models are more efficient in unveiling the inner

Method	Precision	Recall	F1-score	AUC
SE	0.8299	0.8193	0.8203	0.9907
ECA	0.8231	0.8160	0.8153	0.9921
CBAM	0.7977	0.7865	0.7870	0.9920
CA	0.8330	0.8230	0.8232	**0.9923**
FSCA	0.8290	0.8163	0.8178	0.9899
MFS (our work)	**0.8523**	**0.8474**	**0.8474**	0.9921

Table 3. Comparative results between the proposed model using various attention mechanisms and the proposed attention mechanism on the mini-ImageNet dataset. Significant values are in bold.

Method	Precision	Recall	F1-score	AUC
MobileNet v3	0.8299	0.8193	0.8203	0.9907
Desnenet 121	0.7820	0.7733	0.7729	0.9898
ShufflenetV2_X10[41]	0.8151	0.8109	0.8099	0.9935
Resnet34	0.8115	0.8054	0.8057	0.9914
MobileViT	0.8336	0.8207	0.8214	**0.9942**
Our work	**0.8523**	**0.8474**	**0.8474**	0.9921

Table 4. Comparative results of the current state-of-the-art deep learning models and the proposed method on the mini-ImageNet dataset. Significant values are in bold.

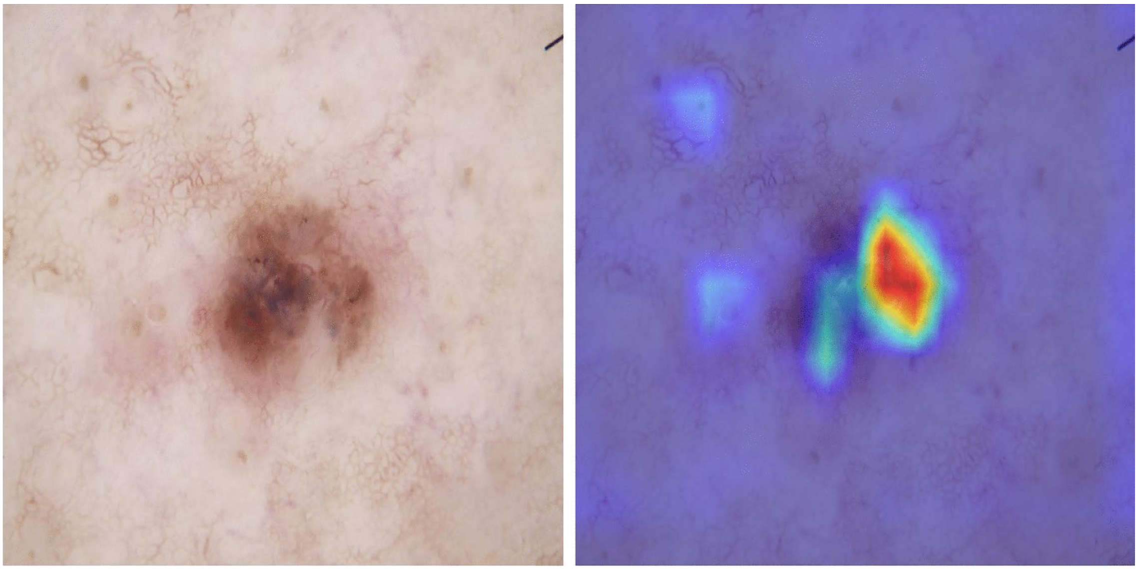

Fig. 7. Example of misclassified image and its corresponding heat map in the ISIC 2019 dataset using the proposed approach.

characteristics to realize accurate image classification. However, the feature extraction of suitable embeddings significantly rely on the adoption of large number of training samples, which were seldom underused in a big portion of the early research. The proposed approach in this work initially integrated the pre-trained models that were trained on a sizable dataset to achieve a high accuracy for skin cancer classification.

Dataset description

The categorization of the ISIC 2019 dataset[32–34] into eight categories has proven to be particularly challenging due to the uneven distribution of images across each category. The following is an outline of the dataset: the total number of images is 25,331, with a resolution of 256×256 and RGB color space. It comprises actinic keratosis (AK) 867, basal cell carcinoma (BCC) 3323, benign keratosis (BKL) 2624, dermatofibroma (DF) 239, melanoma (MEL) 4522, melanocytic nevus (NV) 12,875, squamous cell carcinoma (SCC) 628, and vascular lesion (VASC) 253. Since this dataset was divided into two distinct components: training and testing with training constituting

Category	No. of all samples	No. of training samples	No. of testing samples
AK	867	694	173
BCC	3323	2659	664
BKL	2624	2100	524
DF	239	192	47
MEL	4522	3618	904
NV	12875	10300	2575
SCC	628	503	125
VASC	253	203	50
Totally	25331	20269	5062

Table 5. Detailed distribution of the leveraged dataset in the following experiments.

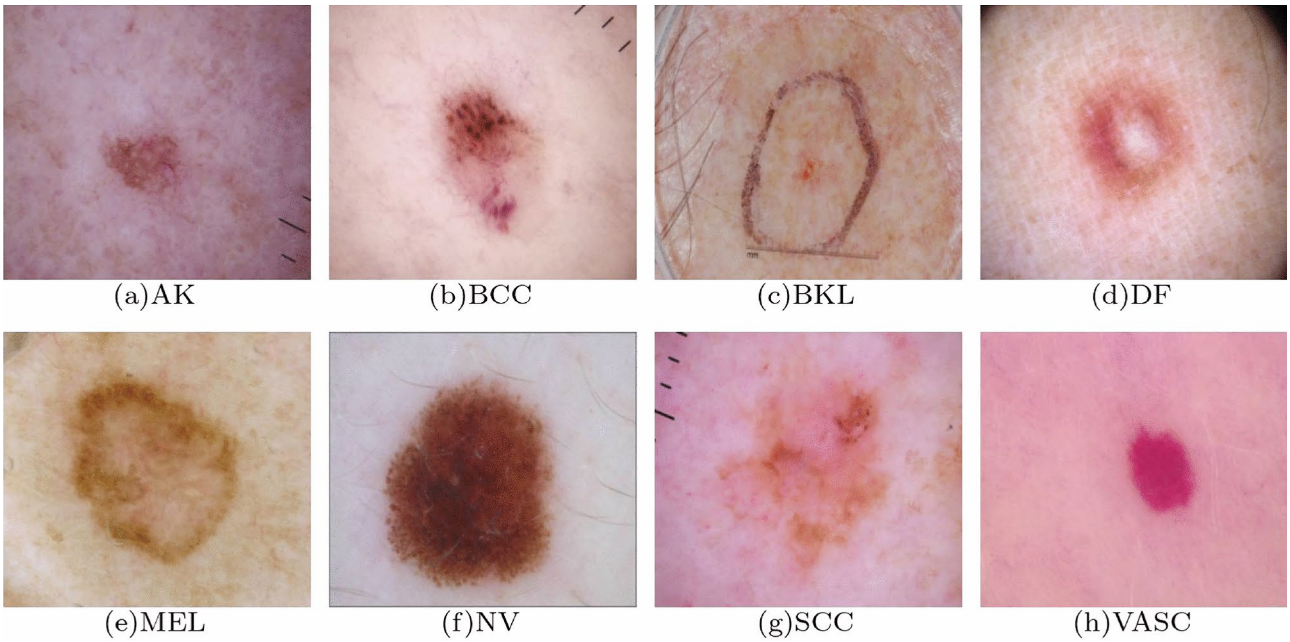

Fig. 8. Image samples in the ISIC 2019 dataset.

80% of the dataset, and testing constituting 20%, the detailed distribution of the leveraged dataset is provided in Table 5. In addition, a group of sample images in this dataset are provided in Fig. 8

Details of the backbone

The configuration of the MobileNet-derived deep learning model is depicted in Fig. 9. The model's design incorporates the primary modules of MobileNet v3, with additional modifications implemented to enhance diagnostic efficacy. The primary component of the proposed model adheres to the structure of MobileNet v3. It encompasses a two-dimensional convolutional layer, a sequence of bottleneck layers of varying dimensions, another two-dimensional convolutional layer, a global average pooling (GAP) layer, and an one-dimensional convolutional layer. This series of modules facilitates the extraction of feature information pertaining to skin lesions, enabling the classification of diseases into distinct kinds using convolutional units. Furthermore, in order to investigate the inclusion of feature information that may not be adequately captured in the original MobileNet v3 model, a multi-scale feature extraction module is employed. In addition, we have put forth a novel attention mechanism as a substitute for the initial attention mechanism module employed in MobileNet v3. As depicted in Fig. 9, the primary module that serves as the bottleneck layer is comprised of an inverted residual network that incorporates depth-wise separated convolutions. The typical convolution procedure was divided into two separate operations: a depth-wise convolution and a point-wise convolution. By employing this approach, the computational complexity and model size are reduced without compromising accuracy. The bottleneck layer incorporates expansion convolution, which primarily aims to augment the channel dimension of the input feature map through the utilisation of a 1×1 convolutional kernel, in conjunction with depth-wise split convolution. The utilization of a 1×1 convolutional kernel is intended to restrict the overall size of the model. In scenarios when the input and output channels exhibit parity, a residual network can be employed. The

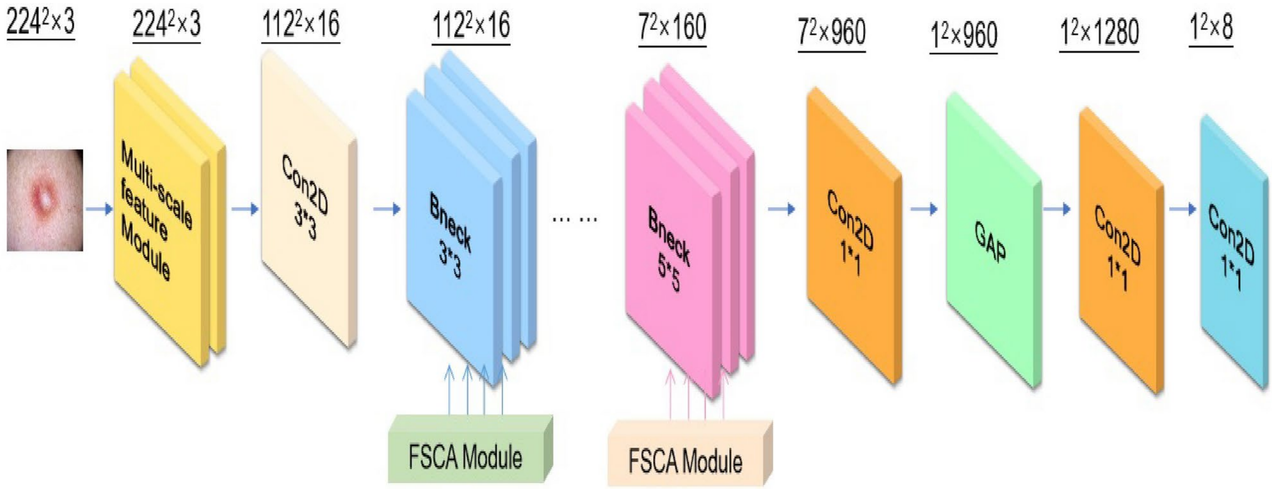

Fig. 9. The underlying structure of the model that was constructed is its core architecture. Bnect is derived from the term bottleneck module, while FSCA refers to the suggested fused spatial-channel attention mechanism. Additionally, GAP indicates the global average pooling unit.

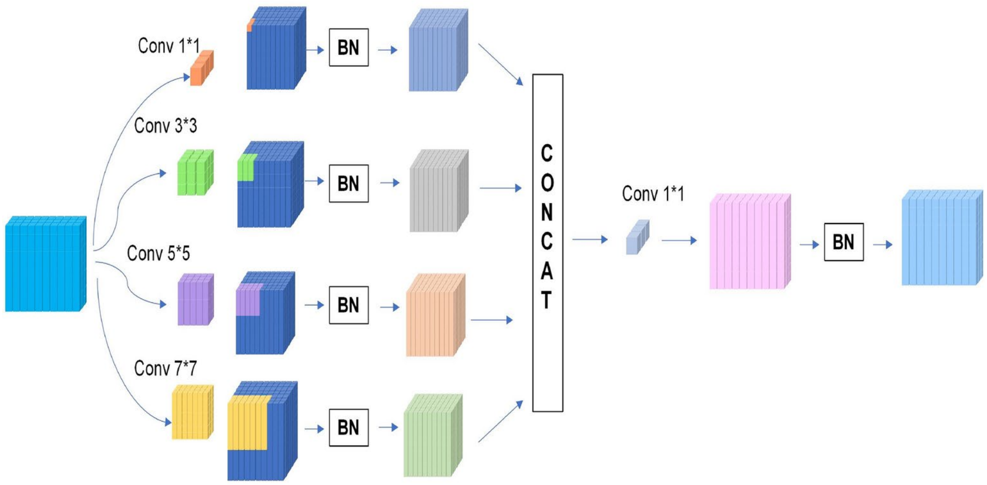

Fig. 10. The multi-scale feature module that is employed to extract the multi-scale feature from an input image. The operation of batch normalization is denoted as BN.

ReLU activation function is utilized to implement the bottleneck layer of the inverted residual structure resulting from the convolution procedures.

Multi-scale feature module

As depicted in Fig. 10, the proposed approach incorporates a multi-scale feature module to extract inner embeddings from several scale levels inside one single skin cancer image. It is worth mentioning that this particular module is well-suited for analyzing skin cancer images due to its ability to extract distinctive features from various kinds of skin lesions at different scales.

Furthermore, the aforementioned characteristics encompass varying dimensions and pose challenges for effective representation using MobileNet V3, which primarily relies on 3×3 and 5×5 convolution operations. In order to enhance the extraction of diverse characteristics across many dimensions, the utilisation of the multi-scale feature module is employed. The comprehensive configuration of this module is illustrated in Fig. 10. The convolutional operators utilized in this study encompass four distinct varieties, namely 1×1, 3×3, 5×5, and 7×7. Following the convolution procedure, the resulting feature maps are subsequently combined into a unified feature vector. By employing this series of processes, it is possible to generate features that incorporate information from various scales of the skin cancer images.

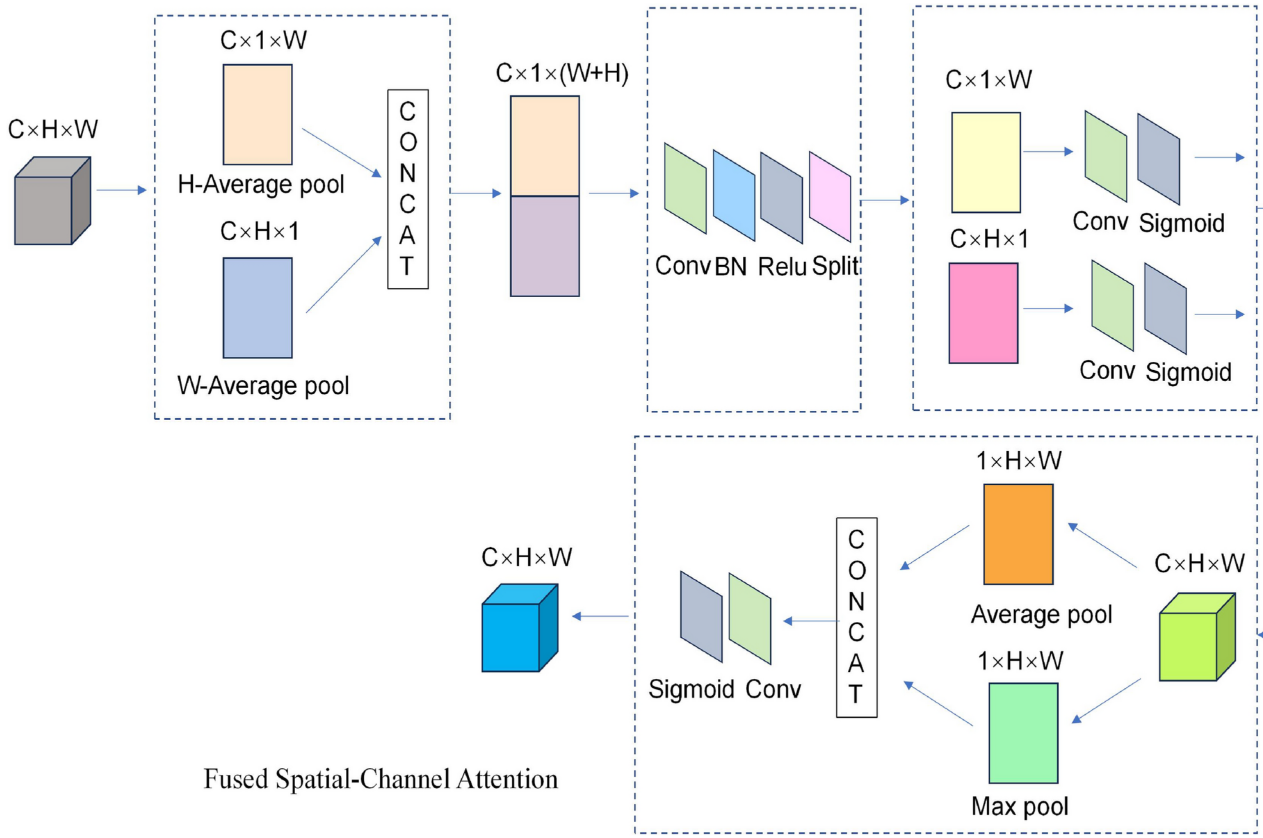

Fig. 11. The depiction of the proposed attention mechanism that is utilized in the suggested method.

Self-attention mechanism

The attention module proposed in this study differs from the original attention mechanism in that it does not rely on global relationship calculations. The architecture of the proposed attention module is depicted in Fig. 11. The MobileNet v3 model incorporated an enhanced version of the squeeze-and-excitation (SE) attention mechanism, as described in the study of Ref.[36]. Nevertheless, the SE module solely considers the relationships among channels, hence disregarding the spatial information. Consequently, the fused spatial-channel attention (FSCA) module was introduced.

The FSCA module takes into account both spatial and channel information from the input layer, resulting in a more efficient guidance of the model towards relevant spots within the image. As depicted in Fig. 11, the FSCA module comprises two successive stages that are concatenated. The initial phase is dedicated to the aggregation of features in both the horizontal and vertical orientations. The output of size $1 \times (H + W) \times C$ is obtained by executing concatenation after averaging pooling in both directions. Subsequently, the weights are applied to the original data in order to generate a collection of directional perception feature layers. The corresponding changes facilitate the attention module in capturing extensive dependencies across spatial directions and retaining positional information, hence aiding in the identification of regions of interest. The subsequent phase of the process is dedicated to channel attention. Within this module, the maximum and average values of the input feature layers can be determined throughout the channels pertaining to each individual feature point. Subsequently, these two values are combined and the channel count is modified through the use of a convolution operation with a single channel. Next, the weights of each feature point in the input feature layer are obtained by using a sigmoid activation function. Ultimately, through the process of combining and multiplying the results obtained from the two stages, we provide the proposed FSCA attention mechanism. This mechanism is designed to concentrate on both the two-dimensional aspects of the input images as well as the channels.

Data availability

The dataset International Skin Imaging Collaboration (ISIC) 2019 used in this study can be downloaded from https://api.isic-archive.com/collections/?pinned=true. The dataset mini-ImageNet used in this study can be downloaded from https://pan.baidu.com/s/1Uro6RuEbRGGCQ8iXvF2SAQ using the password **hl31**.

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H.C.: Writing and Methodology; J.L.: Conceptualization, Methodology, Software, and Writing; W.J.: Software, Administration and Writing. All authors contributed equally to the writing of the manuscript. All the authors consent for publication.

Declarations

Competing interests
The authors declare no competing interests.

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